Dabin Kim

ML / DL Researcher

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Profile

- An ML/DL researcher focused on controllable generative modeling, representation learning, and audio/music understanding. I build systems that translate musical intent and structured conditions into precise control over timbre, pitch, loudness, temporal events, and sound behavior. My work combines deep generative models, differentiable signal processing, and evaluation design, with composer and sound-artist experience serving as a practical lens for musically meaningful model behavior.
- Affiliation: Music and Audio Computing (MAC) Lab, KAIST
- Research Interests: controllable audio generation, music generation and editing, musical representation learning

Education

- Mar 2026 - Present | Ph.D. in Music and Audio Computing (MAC) Lab, KAIST
- Aug 2023 - Aug 2025 | M.S. in Music and Audio Computing (MAC) Lab, KAIST
- Feb 2017 - Feb 2023 | B.S. in Art & Technology, Sogang University; Double major in Big Data Science

Experience

- Jul 2025 - Oct 2025 | Assistant Researcher, Sony Computer Science Laboratories (Sony CSL)

Publications

- AdaTT: Text-Guided Instrument Timbre Transfer with Target-Adaptive Structural Control. Dabin Kim, Junwon Lee, Juhan Nam. 27th Annual Conference of the International Speech Communication Association (INTERSPEECH), 2026. Paper: https://arxiv.org/abs/2606.15813 | Demo: https://dabinkim0.github.io/publications/adatt/
- DDSP-Based Neural Vehicle Sound Synthesis from Driving Signals. Minsuk Choi, Dabin Kim, Daehun Song, Juhan Nam. 6th AES International Conference on Automotive Audio, 2026. Demo: https://dabinkim0.github.io/publications/ddsp-carsound/
- Video-Foley: Two-Stage Video-To-Sound Generation via Temporal Event Condition for Foley Sound. Junwon Lee, Jaekwon Im, Dabin Kim, Juhan Nam. IEEE/ACM Transactions on Audio, Speech, and Language Processing (TASLP), 2025. Paper: https://arxiv.org/abs/2408.11915 | Demo: https://jnwnlee.github.io/video-foley-demo/ | Code: https://github.com/jnwnlee/video-foley
- Pitch-ControlNet: Continuous Pitch Control for Monophonic Instrument Sound Generation. Dabin Kim*, Junwon Lee*, Minseo Kim*, Juhan Nam (*equal contribution). Late-Breaking/Demo (LBD), 25th International Society for Music Information Retrieval Conference (ISMIR), 2024. Paper: https://ismir2024program.ismir.net/lbd_480.html

Selected Presentations

- May 30, 2026 | AdaTT: Text-Guided Instrument Timbre Transfer with Target-Adaptive Structural Control. Short Oral and Poster Sessions, Korean Society for Music Informatics (KSMI) 2026. Jeong Ha Sang Hall, Sogang University, Seoul, South Korea.
- May 12, 2026 | AI generative models for controllable music and Foley sound synthesis. KOBA

2026 AI Audio/Sound Day. COEX Conference Room, Seoul, South Korea.

Skills

- Programming Languages: Python, JavaScript
- ML/DL Frameworks: PyTorch, PyTorch Lightning
- Creative & Audio Tools: Ableton Live, Logic Pro, Max/MSP, SuperCollider